

Embedded System

ALPHA TEAM 9

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Boat Animation on OLED Display using Arduino UNO

Abstract

This experiment demonstrates the implementation of a simple animation (moving boat) on an OLED display using an Arduino UNO. The system uses I2C communication to interface the OLED display and utilizes graphical functions to create and animate shapes. The project helps in understanding embedded graphics, display interfacing, and basic animation logic.

Objective

To design and display a moving boat animation on an OLED screen using Arduino UNO.

Components Required

- Arduino UNO
 - OLED Display (SSD1306, 128×32)
 - Jumper wires
 - USB cable
 - Breadboard (optional)
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Software Requirements

- Arduino IDE
 - Adafruit GFX Library
 - Adafruit SSD1306 Library
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Circuit Description

The OLED display is connected to the Arduino UNO using I2C communication:

- VCC → 5V
- GND → GND
- SDA → A4
- SCL → A5

The display address is set to **0x3C**, which is commonly used for SSD1306 OLED modules.

Working Principle

The OLED display uses the SSD1306 driver, which communicates with the Arduino via I2C protocol. The Arduino sends pixel data to the display using predefined graphics functions.

A boat shape is created using:

- Triangles
- Rectangles
- Lines

To create animation:

- A variable (x) controls horizontal movement
- The boat is redrawn at a new position repeatedly
- When the boat reaches the screen edge, it resets to the starting position

Program Description

The program includes:

- Initialization of OLED display
- Clearing the display buffer
- Drawing shapes to form a boat
- Updating position using a loop
- Delay for smooth animation

Key Logic

```
x = x + 2;  
if(x > 128) {  
    x = -80;  
}
```

This logic moves the boat from left to right continuously.

Coding

```
#include <Wire.h>
```

```
#include <Adafruit_GFX.h>
```

```
#include <Adafruit_SSD1306.h>

#define SCREEN_WIDTH 128
#define SCREEN_HEIGHT 32


Adafruit_SSD1306 display(SCREEN_WIDTH, SCREEN_HEIGHT, &Wire, -1);

void setup() {
  if(!display.begin(SSD1306_SWITCHCAPVCC, 0x3C)) {
    while(1);
  }
  display.clearDisplay();

  // Boat base
  display.fillTriangle(20, 24, 80, 24, 50, 30, WHITE);

  // Boat body
  display.drawRect(30, 20, 40, 5, WHITE);

  // Mast
  display.drawLine(50, 10, 50, 20, WHITE);

  // Sail
  display.fillTriangle(50, 10, 70, 18, 50, 18, WHITE);

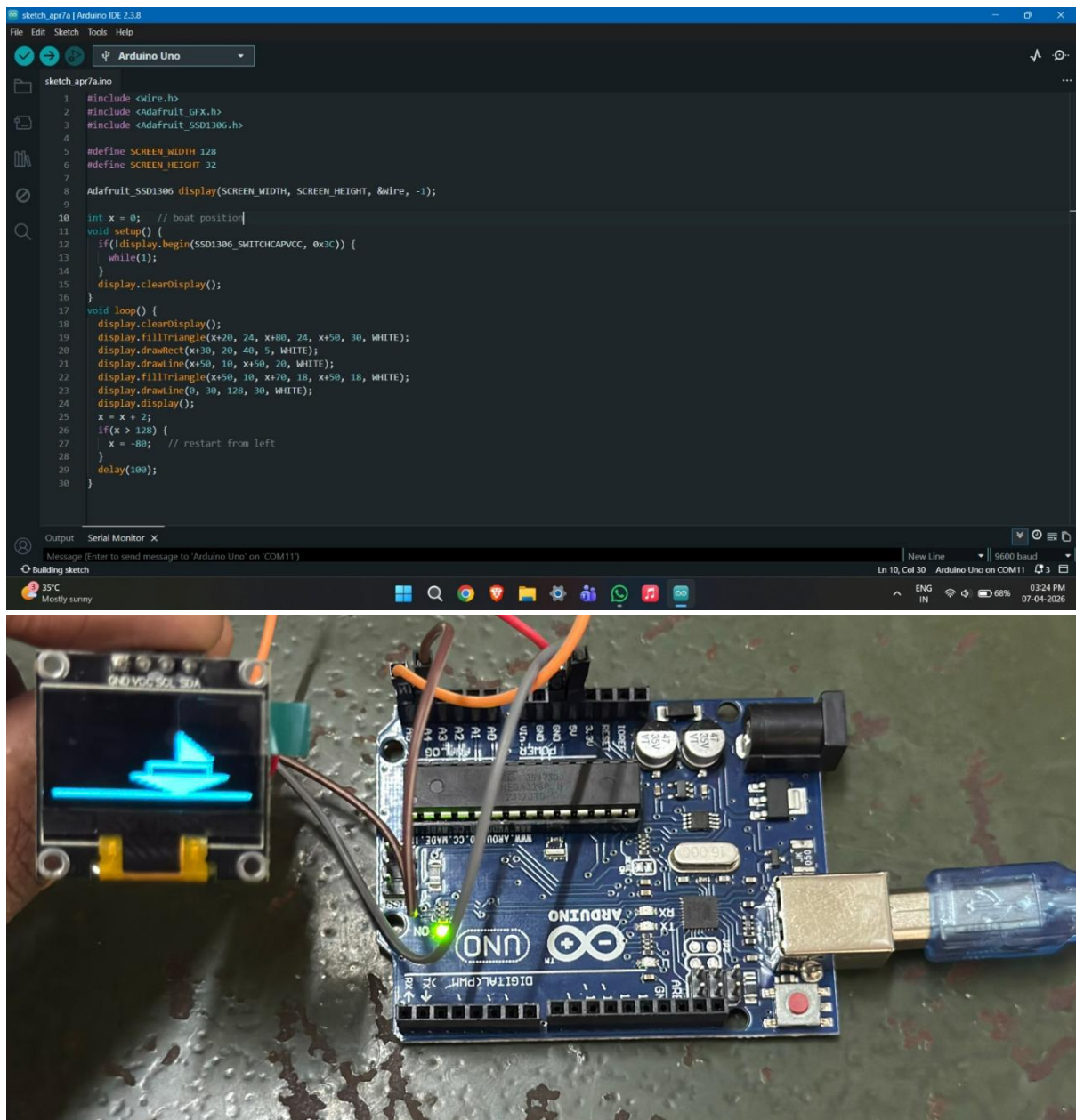
  // Water line
  display.drawLine(0, 30, 128, 30, WHITE);

  display.display();
}

void loop() {

}
```

Image of Coding and Working



Output

- A boat is displayed on the OLED screen
- The boat moves smoothly from left to right
- Once it reaches the end, it restarts from the left side

Applications

- Embedded system animations
- Display interfacing projects

- IoT dashboards
 - Educational demonstrations
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Advantages

- Low power consumption
 - Compact display module
 - Easy interfacing using I2C
 - Smooth graphical rendering
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Limitations

- Limited resolution (128×32)
 - Monochrome display
 - Limited memory for complex graphics
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Conclusion

The experiment successfully demonstrates how to create simple animations using an OLED display and Arduino. It provides practical knowledge of display interfacing and graphical programming, which can be extended to more advanced embedded applications.

Result

The moving boat animation was successfully implemented and displayed on the OLED screen using Arduino UNO.

Object Detection using IR Sensor and OLED Display (Arduino UNO)

Abstract

This experiment demonstrates object detection using an Infrared (IR) sensor and displaying the result on an OLED screen using Arduino UNO. The IR sensor detects the presence of an object based on reflected infrared light, and the OLED display shows messages such as “OBJECT DETECTED” or “NO OBJECT”. This project helps in understanding sensor interfacing and real-time display systems in embedded applications.

Objective

To detect the presence of an object using an IR sensor and display the status on an OLED screen.

Components Required

- Arduino UNO
 - IR Sensor Module
 - OLED Display (SSD1306, 128×32)
 - Jumper wires
 - USB cable
-

Software Requirements

- Arduino IDE
 - Adafruit GFX Library
 - Adafruit SSD1306 Library
-

Circuit Description

IR Sensor Connections:

- VCC → 5V
- GND → GND
- OUT → Digital Pin 2

OLED Display Connections (I2C):

- VCC → 5V
- GND → GND
- SDA → A4
- SCL → A5

The OLED display uses I2C communication with address **0x3C**.

Working Principle

The IR sensor emits infrared light and detects reflections from nearby objects.

- When an object is present → Sensor output = LOW
- When no object is present → Sensor output = HIGH

The Arduino reads the sensor output and displays:

- **“OBJECT DETECTED”** when object is present
- **“NO OBJECT”** when object is absent

The OLED screen updates continuously in real time.

Program Description

The program includes:

- Initialization of OLED display
- Setting IR sensor pin as input
- Reading sensor value using `digitalRead()`
- Displaying text based on sensor condition

Key Logic

```
int sensorValue = digitalRead(IR_SENSOR);
if(sensorValue == LOW) {
    display.println("OBJECT DETECTED");
}
else {
    display.println("NO OBJECT");
}
```

Coding


```

#include <Wire.h>

#include <Adafruit_GFX.h>

#include <Adafruit_SSD1306.h>

#define SCREEN_WIDTH 128

#define SCREEN_HEIGHT 32

#define IR_SENSOR 2

Adafruit_SSD1306 display(SCREEN_WIDTH, SCREEN_HEIGHT, &Wire, -1);

void setup() {
  pinMode(IR_SENSOR, INPUT);

  if(!display.begin(SSD1306_SWITCHCAPVCC, 0x3C)) {
    while(1);
  }

  display.clearDisplay();

  display.setTextSize(2);

  display.setTextColor(WHITE);
}

void loop() {
  int sensorValue = digitalRead(IR_SENSOR);

  display.clearDisplay();

  if(sensorValue == LOW) {
    display.setCursor(0, 10);
    display.println("OBJECT");
    display.setCursor(0, 20);
    display.println("DETECTED");
  }
  else {
    display.setCursor(0, 10);
    display.println("NO OBJECT");
  }
}

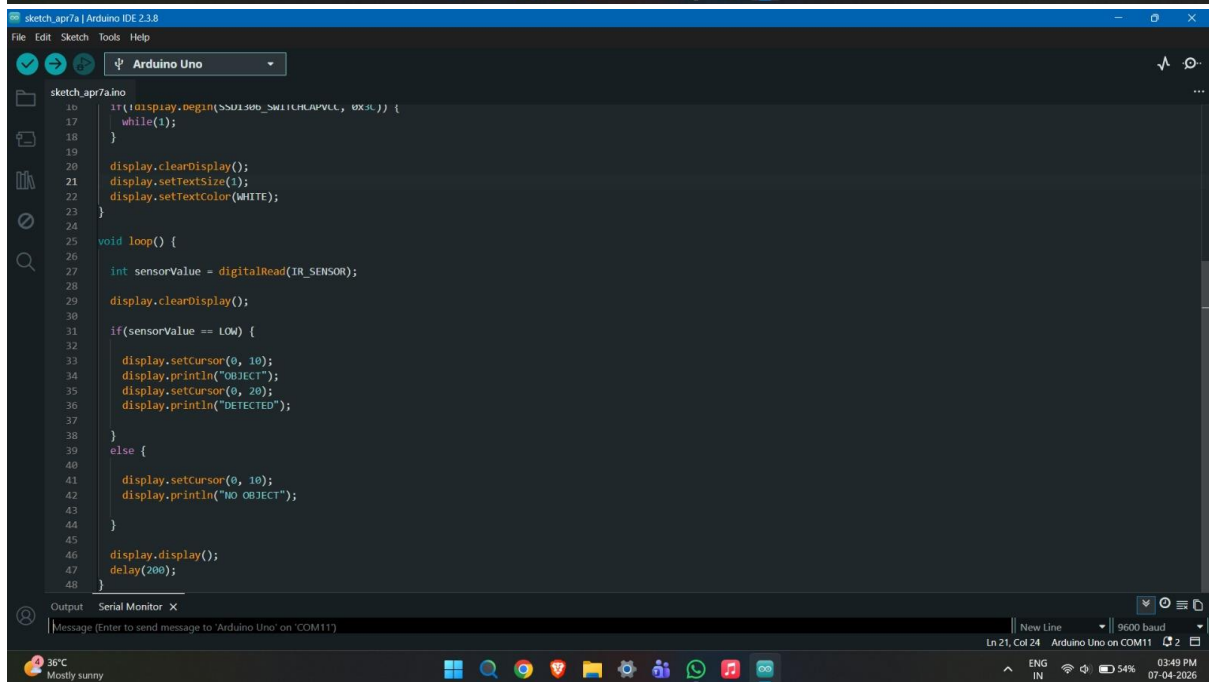
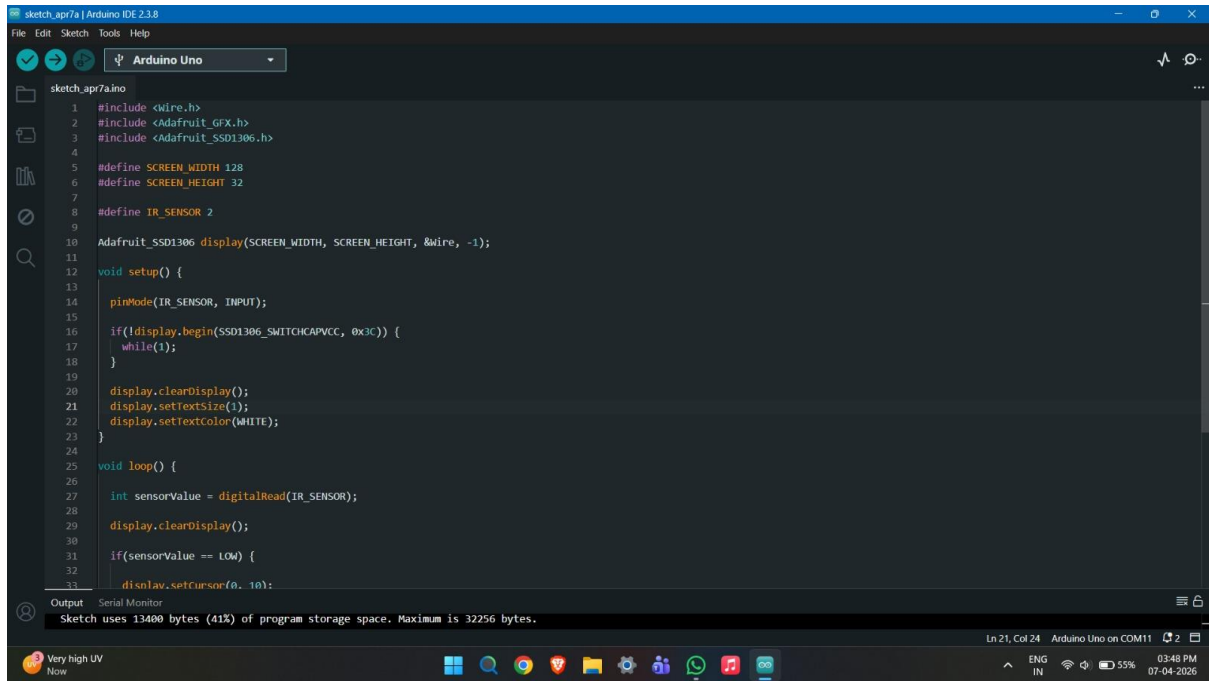
```

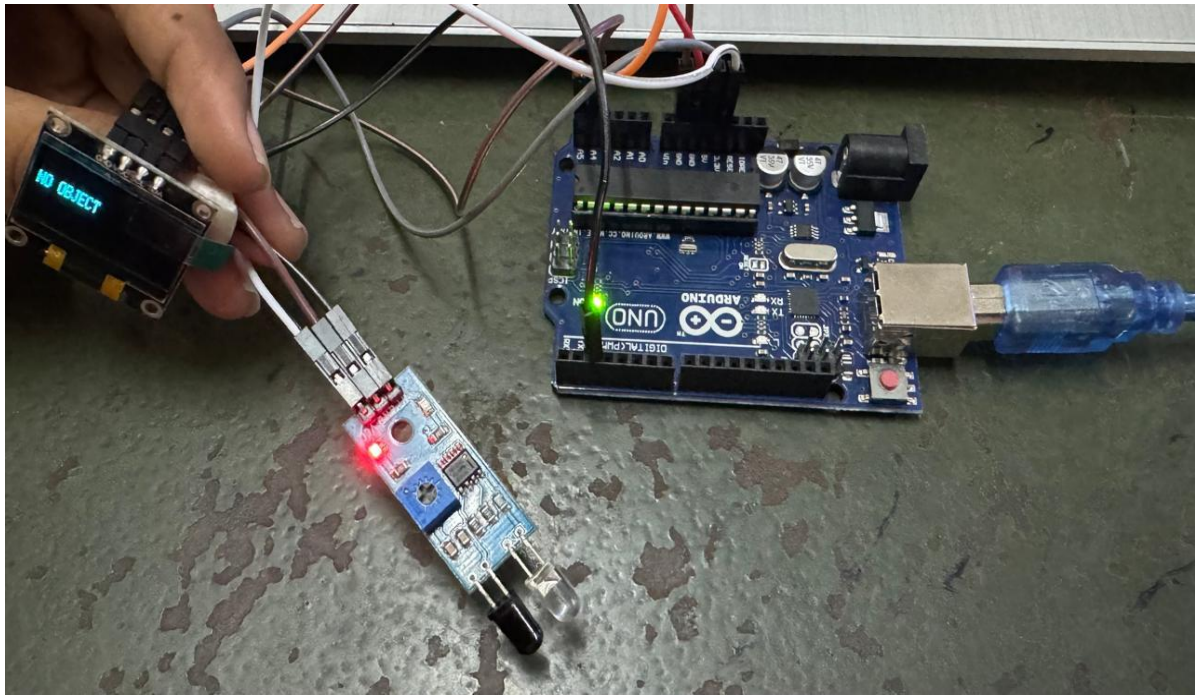
```
display.display();

delay(200);

}
```

Image of Coding and Working





Output

- OLED displays **“NO OBJECT”** when no object is near the sensor
 - OLED displays **“OBJECT DETECTED”** when an object is placed near the sensor
 - Real-time detection and display
-

Applications

- Obstacle detection systems
 - Automatic doors
 - Industrial automation
 - Smart security systems
 - Robotics
-

Advantages

- Simple and low-cost system

- Fast response time
 - Easy interfacing with Arduino
 - Real-time display output
-

Limitations

- Limited detection range
 - Affected by ambient light conditions
 - Cannot detect transparent objects effectively
-

Conclusion

The experiment successfully demonstrates object detection using an IR sensor and displaying the result on an OLED screen. It provides a practical understanding of sensor interfacing and real-time monitoring systems, which can be extended to advanced automation applications.

Result

The system successfully detected the presence and absence of objects and displayed the corresponding message on the OLED display.